

Combustion

Can you guess what is common in rusting of iron, burning coal and the flame of candle? They all undergo a chemical reaction known as combustion. Combustion is a chemical reaction that occurs in the presence of a fuel and an oxidizing agent that produces energy, usually in the form of heat and light.

What we call as 'burning' is really a combustion reaction. In fact combustion is one of the first chemical reactions intentionally harnessed by humans. Any reaction that involves reaction with oxygen is called oxidation reaction. In the combustion of hydrocarbon with oxygen, typically carbon dioxide and water are produced.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + Heat energy

Hydrocarbon + Oxygen → Carbondioxide + Water

All combustion reactions are exothermic; that is they release heat.

Ignition Temperature:

The minimum temperature at which a substance catches fire and burns is called its ignition temperature.

A substance will not catch fire and burn if its temperature is lower than its ignition temperature. Different substances have different ignition temperatures.

Substances which have very low ignition temperature and can easily catch fire with a flame are called inflammable substances. E.g. Petrol. Alcohol, LPG (Liquefied Petroleum Gas), CNG (Compressed Natural Gas), etc.

Flame



Flame is actually a chemical reaction. To be specific, the flame is a mixture of gases (vaporized fuel, oxygen, carbon dioxide, carbon monoxide, water vapor, and many volatile materials) and so is matter. The light and heat produced by the flame is energy, not matter. But fire is a matter.

Flame and its structure

Fact

Chemical Reaction in Ignition

Oxygen + Heat + Fuel → Fire

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